Burak Ömür

Senior AI Engineer

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PROFILE

AI engineer building and running production LLM systems since 2022. Lead architect of an agentic coaching product at Retorio, owning runtime, model routing, memory, evaluation, and observability end to end under EU data residency and GDPR constraints.

EXPERIENCE

Retorio GmbH Munich, DE — Mar 2022 – Present

Senior AI Engineer (Working Student, Mar 2022 – Jun 2024)

- Lead the architecture of **Tori**, an agentic coaching product, as the sole AI engineer: runtime, model routing, state and memory, evaluation, and observability.
- Built the runtime on a self-hosted Agent Protocol server (Aegra) with LangGraph and DeepAgents for long-horizon planning and subagent delegation, with durable event logging (Redis Streams into PostgreSQL) making multi-turn sessions replayable and auditable.
- Route all model traffic through a central LiteLLM proxy on Cloud Run against Vertex AI EU endpoints: Redis-backed fleet-wide admission control, isolated real-time and feedback services, cold starts cut from roughly 60s to negligible.
- Designed a layered evaluation stack: Langfuse for online production scoring, AgentEvals for LangGraph trajectory checks in CI, MLflow for judge-to-human alignment gated on Cohen's $\kappa \geq 0.6$, controlling for self-preference bias.
- Designed per-user memory banks and a source-addressed evidence store with bi-temporal provenance (PROV-O style): grounded feedback with citations and clean GDPR erasure.
- Defined the SLI and measurement design behind a contractual SLA for an enterprise customer, using real user traffic and application-layer error signals to catch mid-stream failures that Cloud Run metrics hide.
- Earlier: FastAPI and gRPC microservices on Docker/GCP, an in-house face detection model and proprietary NLP models, LLM chains with retry and fallback handling.

Tazi.ai Istanbul, TR — Jul 2020 – Sep 2020

Machine Learning Engineer, Intern

- Developed a feature selection method based on variational autoencoders in Scala and Deeplearning4j.

SÜTAŞ Süt Ürünleri A.Ş. Istanbul, TR — Jun 2019 – Aug 2019

Software Engineer, Intern

- Built an internal Android application to shorten communication paths between production hierarchy levels.

TECHNICAL SKILLS

Languages: Python (expert), SQL (expert), Java, C++, Scala

LLM & Agents: LangGraph, LangChain, DeepAgents, Agent Protocol (Aegra), MCP / FastMCP, LiteLLM, RAG

Eval & Observability: Langfuse, AgentEvals, MLflow, DeepEval, OpenTelemetry, Grafana

Backend & Data: FastAPI, gRPC, Pydantic, Docker, Redis, PostgreSQL / Cloud SQL, PgBouncer, MongoDB

ML & Cloud: PyTorch, TensorFlow, JAX, computer vision; GCP (Cloud Run, Cloud SQL, Vertex AI), Azure, CI/CD

Compliance: GDPR, EU AI Act, EU data residency architectures

SELECTED PROJECTS

Towards Smarter and Context-Aware Retrieval Augmented Generation with LLMs *M.Sc. Thesis, TUM*

- Built a query taxonomy for RAG failure modes (feature extraction, disambiguation, hallucination) and tested explicit against implicit feature extraction.
- Created AbbrvQA, an AI-generated ambiguity dataset, and proposed a RAG pipeline with disambiguation and validation modules.

Investigating Bias and Discrimination in Movies Using Deep Learning *B.Sc. Thesis, Boğaziçi*

- Computer vision analysis of the MovieNet dataset using pre-trained architectures in Python. Advisors: Pınar Yanardağ, Suzan Üsküdarlı.

EDUCATION

Technical University of Munich Munich, DE — Oct 2021 – Jun 2024

M.Sc. Informatics (GPA 1.61)

Boğaziçi University Istanbul, TR — Sep 2016 – Jun 2021

B.Sc. Computer Engineering (GPA 3.75/4.00)

Languages: Turkish (native), English (fluent), German (A2)